

A-LEVEL QUALITATIVE QUESTIONS

H2 Physics (Syllabus 9478)

1. READING AND UNDERSTANDING THE QUESTION

- **Identify Key Concepts:** Read carefully to identify specific requirements. Do not regurgitate memorized textbook answers that do not fit the scenario.
- **Look out for Command Words:** Refer to Syllabus 9478 Glossary of Terms (p. 37).

Some examples include:

STATE	DESCRIBE	EXPLAIN
give a concise answer with little or no supporting argument.	state in words (using diagrams where appropriate) the main points of the topic.	may imply reasoning or some reference to theory, depending on the context.

- **Link to Context:** Always refer to given parameters, labels, diagram details, or experimental setups to justify your deductions. The question may be a follow-up question from previous parts.

2. STRUCTURING YOUR ANSWERS

- **Use Physics Keywords:** Explicitly state core physics principles, laws and topic-specific terminology. Avoid using layman terms.
- **Marks Allocation:** Use marks as a guide for number of required points (e.g. 3 marks → 3 points).
- **Formulae Support, Not Replace:** Standard equations (e.g., $F = ma$, $V = IR$) may be quoted without defining standard symbols. However, a purely mathematical derivation is not a substitute for a written explanation.
- **Logical & Concise Flow:** Keep answers legible and succinct.

3. EXAM RULES & MARKING MECHANICS

- **Stick to 'N' Responses:** If question asked for 2 items, write only 2 responses. Extra points will not earn marks. In fact, contradictory answers or wrong answers may invalidate the correct ones. Put your most confident answers first.

✗ Do not use Abbreviations & Symbols for qualitative questions!

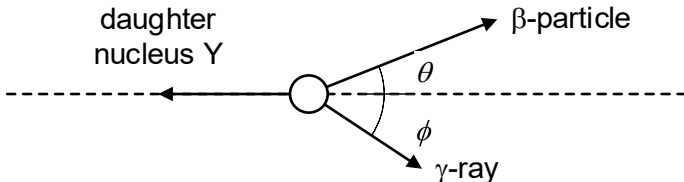
Write terms out in full! Do NOT use: N2L, PCM, PCE, K.E., P.E., w.d., p.d., ↑, ↓

Qualitative Questions from past year A-Level papers

1	2011 Paper 3 Q1(a)	Motion and Forces, Collisions
	A toy car with a rocket engine moves along a horizontal track. The rocket engine produces a constant forward force of 4.6 N. The car loses mass continuously as exhaust gases are produced by the rocket. Use momentum considerations to explain why the rocket produces a forward force on the car. [3]	
	Key Concepts	Conservation of momentum, Newton's laws of motion
	Answer	- As exhaust gas is continuously ejected backward, the momentum of the gas increases in the backward direction. - By Newton's 2nd law, the rate of change of momentum of the gas is equal to the force exerted on the gas by the car in the backward direction. - By Newton's 3rd law, the gas exerts an equal magnitude of force on the car in the forward direction.
	Alternative Answer	- As exhaust gas is continuously ejected backward, the momentum of the gas increases in the backward direction. - Since the total momentum of the exhaust gas-car system must remain constant by the principle of conservation of momentum, the momentum of the car must increase in the forward direction. - By Newton's 2nd law, the rate of change of momentum of the car is equal to the force exerted on the car by the gas.
	Remarks	Mentioning "momentum considerations" and "forward force" point to using PCM and Newton's 2nd law. Note that forward force requires ejecting exhaust gas backwards.

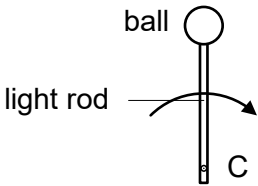
2	2012 Paper 3 Q6(b)(ii)	Collisions
	Two frictionless trolleys A and B are moving horizontally in the same direction. Trolley A has speed 4 m s^{-1} and trolley B has speed 2 m s^{-1}. The trolleys collide. Explain whether, during the collision, it is possible for both trolleys to be at rest simultaneously. [3]	
	Key Concepts	Conservation of momentum
	Answer	- The initial total momentum of the system of trolleys is non-zero. - Since no resultant external force acts on the system, by the principle of conservation of momentum, the total momentum of the system must remain non-zero. Therefore, it is not possible for both trolleys to be at rest simultaneously as that will mean that the total momentum is zero.
	Remarks	Explain either that zero total momentum violates PCM, or that satisfying PCM requires non-zero total momentum.

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3	2012 Paper 3 Q6(d)(ii)	Collisions
*	The daughter nucleus Y may move after the decay of a stationary parent nucleus. Possible directions for the β-particle, the γ-ray photon and the daughter nucleus Y are shown below.  Momentum is a vector quantity and can be resolved into perpendicular components. Suggest and explain the relation between the angle θ and the angle ϕ. [3]	
	Key Concepts	Conservation of momentum
	Answer	- The parent nucleus is initially at rest i.e., initial momentum is zero. By the principle of conservation of momentum, the total momentum of daughter nucleus Y, β-particle and γ-ray must be zero. - Since the daughter nucleus recoils horizontally, the vertical component of the momentum of the β-particle (p_β) is equal in magnitude but opposite in direction to that of the γ-ray (p_γ). - Therefore, $p_\beta \sin \theta = p_\gamma \sin \phi$
	Remarks	Since the parent nucleus is initially at rest, by the principle of conservation of momentum, the total momentum of daughter nucleus Y, β -particle and γ -ray in the horizontal direction must also be zero.

4	2015 Paper 3 Q1(b)(ii)	Forces and Moments, Circular Motion, Gravitational Fields
	State and explain whether a satellite in a circular Earth orbit is in equilibrium. [3]	
	Key Concepts	Newton's laws of motion, circular motion, gravitation
	Answer	- For the satellite to perform circular motion around the Earth, the satellite must experience a centripetal force (or centripetal acceleration) towards the centre of the Earth. - The centripetal force (or acceleration) is provided by the gravitational force exerted on the satellite by the Earth. - Since the resultant force on the satellite is not zero, the satellite is not in equilibrium.
	Alternative Answer	- For a satellite to be in equilibrium, the resultant force acting on it must be zero. - According to Newton's 1st law, the satellite must either be at rest or move with constant speed in a straight line. - Since a satellite orbiting the Earth moves in a circular path, the satellite is not in equilibrium.
	Remarks	The condition for equilibrium must be stated.

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5	2010 Paper 3 Q2(c)(i)	Energy and Fields
*	A small ball of mass m is fixed to one end of a light rigid rod. The ball is made to move at constant speed around the circumference of a vertical circle with centre at C, as shown.  Explain why work has to be done for the ball to move from the position where it is vertically above point C to the position where it is vertically below C. [3]	
	Key Concepts	Conservation of energy, circular motion
	Answer	- When the ball moves down to the position vertically below point C, its gravitational potential energy decreases. - However, since the ball moves at constant speed, its kinetic energy remains constant. - By the principle of conservation of energy, there must be negative work done by an external force (a motor exerting an anticlockwise torque at C or a hand supporting the ball or rod as it is lowered) to account for the lower total energy of the ball.
	Remarks	If there is no negative work done by external force, the ball would gain kinetic energy as it loses gravitational potential energy.

6	2011 Paper 3 Q6(d)	Gravitational Fields
*	The satellite in (c) gradually loses energy due to small resistive forces. Suggest why many such satellites eventually 'burn up' in the Earth's atmosphere. [4]	
	Key Concepts	Kinetic energy and total energy of satellite, conservation of energy, air resistance
	Answer	- As the <u>total energy of a satellite</u> $E_T = -GMm/(2R)$ <u>gradually decreases</u>, <u>its orbital radius R decreases</u>. - The speed of the satellite increases as the <u>kinetic energy of a satellite</u> $E_K = GMm/(2R)$ <u>increases with decreasing R</u>. - At a lower R, the atmosphere is denser (or has more air molecules). Coupled with the increase in speed, <u>the air resistance acting on the satellite increases significantly</u>. - Hence, the (negative) rate of work done by air resistance on the satellite increases greatly and <u>results in a large amount of thermal energy being produced</u>, which burns up the satellite.
	Remarks	Essential to explain how "small resistive forces" eventually become significant due to increase in speed at lower altitude. Note: The air resistance f_d acting on a body moving with speed v through air of density ρ is $\frac{1}{2}c_d\rho Av^2$, where c_d is the drag coefficient and A is the cross-sectional area normal to the velocity.

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7	2015 Paper 3 Q6(a)(ii)	Gravitational Fields
	Explain why the acceleration of free fall near the Earth's surface is constant. [3]	
	Key Concepts	Gravitational field strength, acceleration of free fall, centripetal acceleration
	Answer	- For points near surface of Earth, the distance from the centre of Earth does not change significantly due to Earth's large radius. - The gravitational field strength g at a point is inversely proportional to the square of its distance from centre of Earth. So, the magnitude of g is constant for points near surface of Earth. - Since the centripetal acceleration (due to Earth rotation about its axis) is negligible (compared to g), the acceleration of free fall at a point is numerically equal to g at that point. Hence, acceleration of free fall is constant near Earth's surface.
	Remarks	The relationship between acceleration of free fall and gravitational field strength must be made clear.

8	2021 Paper 3 Q2(a)	Gravitational Fields
	The gravitational potential ϕ at a distance x from a point mass M is given by the equation: $\phi = -\frac{GM}{x}$ where G is the gravitational constant. Explain why gravitational potential is a negative quantity. [3]	
	Key Concepts	Gravitational potential, conservation of energy
	Answer	- Gravitational potential at infinity is zero. - Since gravitational force is attractive in nature, to bring a mass from infinity to a point in the gravitational field, the direction of the external force is opposite to the direction of displacement of the mass. - This results in negative work done per unit mass by the external force.
	Remarks	There must be clear explanation of why the work done per unit mass by external force is negative, and to link this work done to gravitational potential via its definition.

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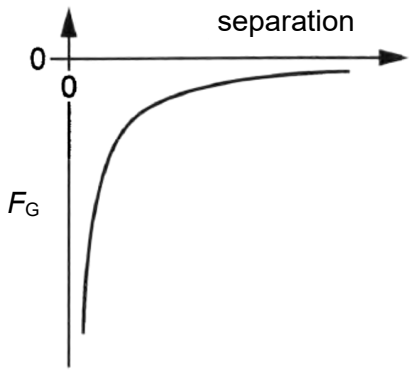
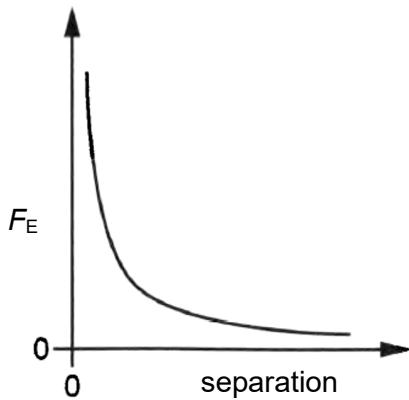
9	2015 Paper 3 Q2(b)	Superposition
	A student determines the wavelength of the light from a laser using a double slit. The laser beam is then adjusted so that the intensity of the light passing through one of the two slits is reduced. Describe the changes, if any, in appearance of the fringes on the screen. [3]	
	Key Concepts	Two-source interference
	Answer	- At positions of constructive interference (bright fringes), <u>the bright fringes become less bright</u>. - At the positions of destructive interference (dark fringes), <u>the dark fringes have non-zero intensity</u> (which are lower than the bright fringes). - Hence, the <u>contrast</u> of the fringe pattern will be <u>poorer</u> (compared to the original).
	Remarks	For clarity, it is advisable to discuss the bright and dark fringes separately. If question requires you to explain, the reason would be: - The light passing through the two slits have unequal amplitude as the intensities are not the same. - At positions of constructive interference (bright fringes), the resultant amplitude will be smaller than before due to the decreased amplitude of light from one of the slits. Hence, the bright fringes become less bright. - At the positions of destructive interference (dark fringes), there will not be complete cancellation due to the unequal amplitudes of the light from the two slits. Hence, the dark fringes have non-zero intensity (which are lower than the bright fringes). - Hence, the contrast of the fringe pattern will be poorer (compared to the original).

10	2017 Paper 2 Q3(c)(iii)	Superposition
	Two radio transmitters, A and B, of the same power are separated by a distance of 5.0 km. They each transmit a radio wave of frequency 1200 kHz. The two waves emitted are in phase at the transmitters. The waves interfere. Suggest a change that can be made at the transmitters that will prevent interference between the two waves without changing the frequency. [2]	
	Key Concepts	Two-source interference
	Answer	- Polarise the two waves - such that the axes of polarisation are perpendicular to each other.
	Remarks	Deduce from the conditions for observable interference what needs to be done to “prevent interference”. Decrease the amplitude of one wave to zero is not a good answer as it defeats the purpose of having two radio transmitters. Making the two waves incoherent to prevent interference is not allowed as the question states that the waves are in phase.

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11	2022 Paper 2 Q3(a)(i)	Electric Fields
	Two charged particles, P and R, are isolated in space and separated by a distance of 6.0 cm as shown below. Particle P has a charge of +6.5 nC and particle R has a charge of -0.45 nC. Explain whether the electric field strength is zero at any point along a straight line between the two charges. [1]	
	Key Concepts	Electric field strength
	Answer	Between the two charges, the electric field due to P is to the right (direction of electric field is away from positive charges) as P is positively charged and the electric field due to R is also to the right (direction of electric field is towards negative charges) as R is negatively charged. Hence, the electric field is not zero at any point.
	Remarks	Focus only on the region between the two charges and be specific about the direction of the electric field due to each charge, i.e., to the right or left, instead of both pointing in the same direction.

12	2022 Paper 2 Q3(a)	Projectile Motion, Gravitational Fields, Electric Fields
	Figure below shows the path of a moving object as it enters and travels through a uniform field of force. Before the object enters this field, its direction of motion is perpendicular to the direction of the field. The object could be a mass in a gravitational field or a charged particle in an electric field. In both cases, the shape of the path is the same. Explain why the shape of the path is the same in both cases. [2]	
	Key Concepts	Gravitational and electric fields, parabolic path
	Answer	- A mass in a gravitational field or a charged particle in an electric field experiences constant force (in magnitude and direction) and therefore undergoes constant acceleration. - Since both have a constant velocity in the direction perpendicular to the acceleration, the path is parabolic.
	Remarks	Keywords such as 'constant force' and 'constant velocity perpendicular to acceleration' must be mentioned. Explanation must refer to both the mass and the charged particle.

13	2022 Paper 2 Q3(b)(i)	Gravitational Fields, Electric Fields
*	The variations with separation of the gravitational force F_G and of the electric force F_E between two protons are shown in the figure below.   Explain why the gravitational force and the electric force have opposite signs. [1]	
	Key Concepts	Gravitational and electric forces
	Answer	Gravitational force between masses is attractive (and is negative), while electric force between like-charges is repulsive (and is positive), therefore the forces are of opposite signs.
	Remarks	Highlight the difference between the two forces. Do not just quote the equations.

14	2022 Paper 2 Q3(b)(ii)	Gravitational Fields, Electric Fields
*	Refer to the previous question for the scenario. State and explain whether the axes on both graphs are drawn to the same scale. [2]	
	Key Concepts	Gravitational and electric forces
	Answer	No. For the same separation, the magnitude of gravitational force is much smaller than that of the electric force.
	Alternative Answer	No. For the same magnitude of force, the separation (between the two protons) is much larger for the electric force than the gravitational force.
	Remarks	A correct statement with an incorrect explanation will result in zero mark.

15	2020 Paper 3 Q2(a)	Temperature and Ideal Gases
	A sample of an ideal monatomic gas has a volume of $4.2 \times 10^{-3} \text{ m}^3$ at a pressure of $3.6 \times 10^5 \text{ Pa}$ and a temperature of 70°C. Explain why the mean velocity of the atoms of the gas is zero. [2]	
	Key Concepts	Kinetic theory of gas
	Answer	- The atoms of the gas are moving randomly in all directions and with a range of speeds. - Hence, the vector sum of the velocities of all the atoms is zero, which leads to an average velocity of zero.

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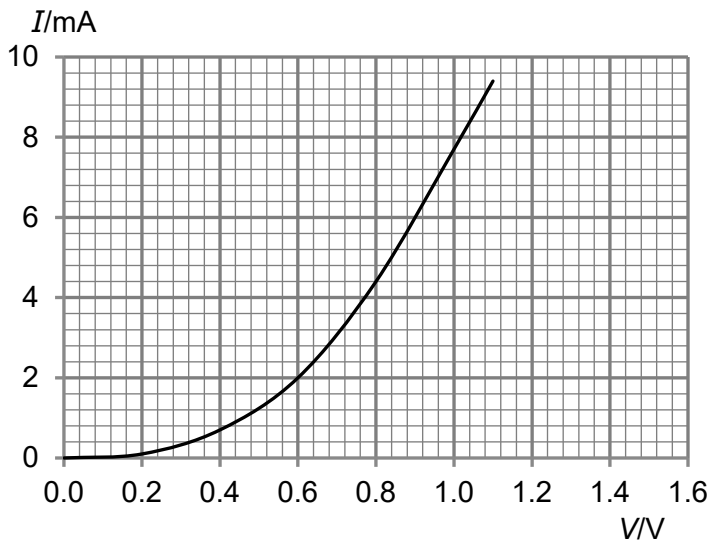
	Remarks	The keywords are “moving randomly in all directions” and “velocity is a vector”. One can also argue that the centre of mass of the atoms remains stationary (else the gas will be moving). Hence, the sum of momenta of all atoms must be zero. Since the atoms have identical mass, the sum of velocities and therefore the mean velocity of all atoms must be zero.
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16	2012 Paper 2 Q4(b)(i)	Temperature and Ideal Gases
	A gas in a cylinder is heated and expands under constant pressure. Explain, in terms of the force produced by the molecules of the gas, how the pressure remains constant as the volume increases. [3]	
	Key Concepts	Kinetic theory of gas, Newton’s laws of motion
	Answer	- When the gas is heated, its temperature increases, causing the average kinetic energy and speed of the gas molecules to increase. Hence, the rate of change of momentum of the molecules upon hitting the walls increases and molecules collide more forcefully with the walls. - When the volume of the gas increases to maintain a constant pressure, the molecules have to travel longer distances between collisions with the walls. Hence, the rate of collisions with the walls decrease and the molecules collide less frequently with the walls. - These two effects negate each other, and the force exerted by the molecules on the walls remains constant. Hence, the pressure remains constant.
	Remarks	This question asks for explanation in terms of force and hence the ideal gas equation cannot be used.

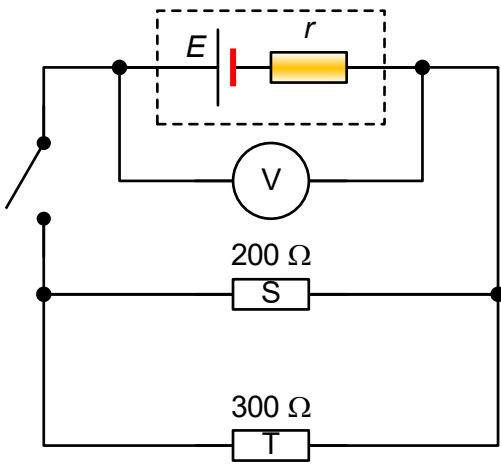
17	2015 Paper 3 Q7(d)(ii)	Thermodynamic System
*	Use the first law of thermodynamics to explain why the specific heat capacity of an ideal gas measured at constant volume is different from the specific heat capacity when measured at constant pressure. [3]	
	Key Concepts	First law of thermodynamics
	Answer	<u>For a gas heated at constant pressure, it expands. W is negative, from $\Delta U = Q_p + W_p$ $\therefore Q_p > \Delta U$.</u> <u>For a gas heated at constant volume, W is zero hence $Q_v = \Delta U$.</u> <u>With the same unit rise in temperature, both processes have the same increase in internal energy, the specific heat capacity at constant pressure is higher than that at constant volume.</u>
	Remarks	*(a)(i) already asked to state Specific Heat Capacity. Hence you can link S.H.C to Q without explaining again. *(d)(i) already asked to state the First law of thermodynamics. Hence you just need to make references to the law.

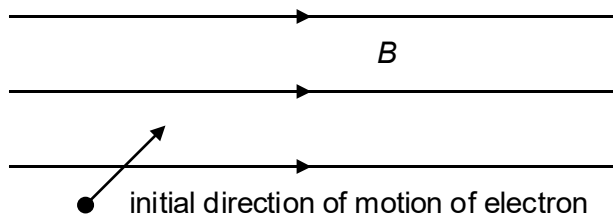
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18	2021 Paper 2 Q6(a)(iii)	Circuits
*	Explain why the current I in the filament lamp is greatest when the current is first switched on. [2]	
	Key Concepts	Variation of resistance of metal with temperature
	Answer	- The current will heat up the filament lamp and causes its temperature to <u>increase with time</u>. - Since <u>the resistance of the filament lamp increases with temperature</u>, its <u>resistance is the lowest when it is first turned on</u> as temperature is the lowest, hence the current is the greatest.
	Remarks	Temperature of the lamp increases with time due to Joule heating. From the electrical characteristic of a tungsten filament, deduce that resistance is lowest at the start.

19	2012 Paper 2 Q3(a)(i)	Circuits
	The variation with potential difference V of current I for a semiconductor diode is shown.  Describe the main features of the figure that show the characteristics of the diode in terms of current, voltage and resistance. [4]	
	Key Concepts	I – V graph of diode
	Answer	<u>Resistance is the ratio of p.d. to current.</u> - When <u>potential difference (p.d.) across the diode is less than 0.12 V</u>, there is <u>negligible current</u> in the diode. Hence, <u>the resistance of the diode is very high at low p.d.</u> <u>Between $V = 0.12\text{ V}$ and 0.80 V, current increases at an increasing rate.</u> <u>the resistance of the diode decreases</u> with increasing V. <u>Beyond 0.80 V, the current increases linearly</u> with p.d. <u>Resistance of the diode continues to decrease</u> with increasing V.
	Remarks	Describe the changes in current and resistance in three distinct voltage ranges.

20	2014 Paper 2 Q3(a)	Circuits
	A battery B, a variable resistor R and a uniform resistance wire PQ are connected in series as shown. The resistance of R is varied from 0 to 12 Ω . Describe and explain the variation in the terminal p.d. across B. [3]	
	Key Concepts	Terminal p.d. of battery
	Answer	- As the resistance of R increases, the current I supplied by battery of e.m.f. E decreases. - The potential difference across the internal resistance r decreases. - Since terminal p.d. $V = E - Ir$, V increases as resistance of R increases.
	Remarks	It is also possible to deduce the variation in V using the potential divider rule: $V = \left(\frac{R + R_{\text{wire}}}{r + R + R_{\text{wire}}} \right) E$ $= \left(1 - \frac{r}{r + R + R_{\text{wire}}} \right) E$ where R is the resistance of R and R_{wire} is the resistance of the uniform resistance wire.

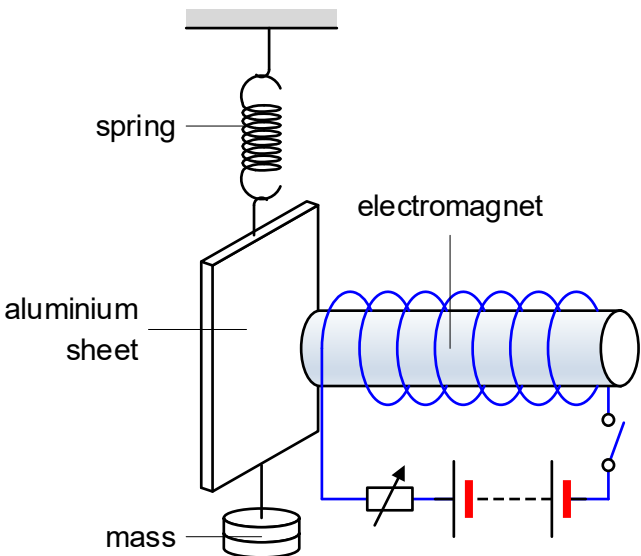
21	2019 Paper 2 Q5(b)(iii)	Circuits
	A power supply of e.m.f. E and internal resistance r is connected to two resistors S and T and a voltmeter.  Resistor T is replaced by an NTC thermistor. Initially the resistance of the thermistor is $300\ \Omega$. The switch is closed. The temperature of the thermistor is then increased. State and explain the change, if any, to the voltmeter reading when the switch is closed. [3]	
	Key Concepts	Resistance in parallel, variation of resistance of thermistor with temperature
	Answer	- As the temperature of the thermistor increases, its resistance decreases. Hence, the equivalent resistance of S and thermistor decreases. - This causes the current I supplied by battery to increase and the p.d. Ir across the internal resistance r to increase. - Since the terminal p.d. is given by $V = E - Ir$, the initial voltmeter reading of E decreases when the switch is closed
	Remarks	This is a variation of Q20.

22	2014 Paper 3 Q7(e)	Electromagnetic Forces
*	An electron enters a uniform magnetic field at an angle as shown below.  By considering the components of the velocity parallel to the magnetic field and at right-angles to the magnetic field, describe and explain qualitatively the motion of the electron in the field. [3]	
	Key Concepts	Magnetic force on charged particle, helical motion
	Answer	The velocity component perpendicular to the magnetic field $v_{\perp}$ produces a magnetic force on the electron that is always perpendicular to $v_{\perp}$ and the magnetic field. This magnetic force on the electron is the centripetal

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		force that causes the electron to perform uniform circular motion in a plane normal to the magnetic field. • The velocity component parallel to the magnetic field produces no magnetic force, but it causes the electron to move at a constant velocity along the magnetic field. • The resultant path of the electron is a helix.
	Remarks	For clarity, discuss the effects of the perpendicular and parallel component of the velocity on the motion of the electron separately.

23	2016 Paper 3 Q4(a)	Electromagnetic Forces
	A positively charged particle, travelling in a vacuum, enters a region of uniform magnetic field. Initially, the path of the particle is normal to the direction of the magnetic field. Explain why the path of the particle in the magnetic field is the arc of a circle. [3]	
	Key Concepts	Magnetic force on charged particle, circular motion
	Answer	• The particle experiences a magnetic force that is always perpendicular to its velocity. • The (radial) magnetic force changes the direction of motion of the particle, but not its speed as there is no (tangential) force parallel to velocity. • Hence the magnitude of the magnetic force ($F = Bqv$) remains constant. • This means magnetic force of constant magnitude provides the centripetal force for the particle to move in an arc of constant radius at constant speed.
	Remarks	One can argue that the kinetic energy and speed of the particle remains constant as the radial magnetic force does zero work on the particle. The terms “radial” and “tangential” are useful in conveying directions. “An arc of constant radius at constant speed” implies that the particle moves in uniform circular motion.

24	2011 Paper 3 Q7(c)(ii)	Electromagnetic Forces
	A spring hangs vertically from a fixed point. An aluminium sheet is attached to the free end of the spring and oscillating vertically. When the current in the coil is switched on, the oscillation of the plate is observed to be damped.  Use Faraday's Law to explain why the oscillations are damped. [3]	
	Key Concepts	Laws of electromagnetic induction, damping
	Answer	- When the current is switched on, the magnetic field produced by the electromagnet links with the aluminium sheet. As the aluminium sheet oscillates vertically, its magnetic flux linkage changes continuously. By Faraday's Law, an e.m.f. is induced within the aluminium sheet. - By Lenz's law, the direction of induced e.m.f. is such that it drives an induced current (eddy current) to oppose the changes in magnetic flux linkage. - The induced current in the sheet produces a retarding force that always opposes its motion, gradually reducing its amplitude.
	Alternative Answer	- When the current is switched on, the magnetic field produced by the electromagnet links with the aluminium sheet. As the aluminium sheet oscillates vertically, its magnetic flux linkage changes continuously. By Faraday's Law, an e.m.f. is induced within the aluminium sheet. - The induced e.m.f. drives a current (eddy current) within the sheet. The energy of the oscillation is gradually converted to electrical energy (the current) which is eventually dissipated as heat in the sheet through Joule heating. - Since the energy of oscillations is proportional to the square of the amplitude, the amplitude of oscillation gradually decreases.
	Remarks	Answers involving EMI usually trend along the following arguments: changing (increasing, decreasing or alternating) magnetic flux linkage, e.m.f. induced using Faraday's law, direction of e.m.f. drives induced current to oppose the change using Lenz's law. Thereafter, one can either use retarding force or energy consideration to explain the phenomenon.

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25	2011 Paper 3 Q7(d)	Electromagnetic Forces
*	Refer to the previous question for the scenario. State and explain the effect on the oscillations in (c) of inserting an iron core into the coil. [3]	
	Key Concepts	Permeability of iron, laws of electromagnetic induction
	Answer	- The <u>iron core increases the magnetic flux density</u> produced by the electromagnet due to its high permeability. <i>(the core would 'concentrate' the magnetic flux is not accepted. It should be appreciated that the core gives rise to an increase flux density over a large area rather than concentrating the existing flux into a small area)</i> As the aluminium sheet oscillates, <u>there is a greater rate of change of magnetic flux linkage which in turn induces a greater e.m.f.</u> - Consequently, <u>a larger induced current flows in the sheet which increases the retarding force acting on the sheet.</u> - This <u>increases the damping effect. Hence, the oscillations will die off more quickly.</u>
	Remarks	In terms of energy, a larger induced current suggests a faster rate of conversion of mechanical energy to electrical energy.

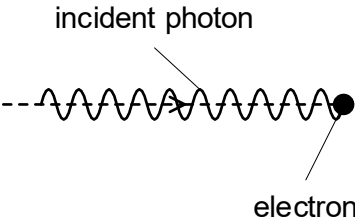
26	2009 Paper 3 Q5(d)	Quantum Physics
	A stream of protons is accelerating towards the star. Suggest why this stream may be a source of X-ray radiation. [3]	
	Key Concepts	X-ray production by accelerating charges
	Answer	- The gravitational field strength around the star is very large due to its large mass. - Hence, the protons experience a large acceleration towards the star. - When charged particles, like protons, undergo large acceleration, they emit high energy photons such as X-rays.
	Remarks	Your answer should address two points: - the high acceleration of the protons, - accelerating charged particles emit electromagnetic radiation – high acceleration / deceleration produces high energy photons

27	2013 Paper 2 Q6(b)	Quantum Physics
*	Use the theory of the particulate nature of electromagnetic radiation to explain why there is a threshold frequency for the photoelectric effect. [3]	
	Key Concepts	Energy of photon, photoelectric equation
	Answer	- According to the particulate theory of light, <u>electromagnetic radiation is made up of discrete quanta of energy known as photons where the energy of each photon is hf (h is the Planck constant and f is the frequency of the radiation).</u> - Since <u>an electron can only absorb one photon,</u>

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		for photoemission to occur, <u>the energy of each photon $hf \geq \Phi$ must be greater than the minimum energy to eject an electron from the surface of a metal (or work function of the metal)</u>. Hence a threshold frequency $f = \Phi/h$ is required for photoemission to take place.
	Remarks	“An electron can only absorb one photon” is the key point. Without which, threshold frequency does not exist. Use key words such as “photons” and “work function” to help convey your understanding of the phenomenon.

28	2020 Paper 2 Q7(c)(iii)	Quantum Physics
	The energy E in eV of each energy level of the hydrogen atom may be determined using the expression $E = -\frac{13.6}{n^2}$ where n is the energy level. Explain why the energy of each energy level is negative. [1]	
	Key Concepts	Discrete energy level of atom
	Answer	The <u>electron in the atom is bound to the nucleus by an attractive Coulombic force</u>.
	Remarks	Key words such as “bound system” and “attractive force” are sufficient to earn a mark. If more marks were allocated, it may be necessary to discuss the work done by an external force to separate the constituents to infinity (where potential energy is zero), and the positive work done by external this force. This positive work done raises the energy of an atom to zero ($n \rightarrow \infty$) and from this, we can deduce that the initial energy of the atom must be negative.

29	2021 Paper 2 Q8(d)(ii)	Quantum Physics
	The Compton effect involves the scattering of an incident X-ray photon.  State and explain the change, if any, of the wavelength of the X-ray photon due to scattering. [1]	
	Key Concepts	De Broglie equation, conservation of energy, energy of photon
	Answer	The incident X-ray photon loses some energy to the electron during the scattering process. Since energy E of photon $E = hc/\lambda$, the scattered X-ray photon with a lower energy has a longer wavelength λ.
	Remarks	An earlier paragraph stated that the incident photon gave energy to an electron during the process of Compton scattering.

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		Note: The incident photon behaves like a particle as it collides with an electron. The scattering process is similar to a basketball colliding with a tennis ball.
30	2010 Paper 3 Q8(d)(ii)	Nuclear Physics
*	Initially, a radioactive source contains N nuclei of bismuth-212. After two hours, it is found that the number of bismuth-212 nuclei has reduced to approximately $\frac{1}{4}N$. However, although bismuth-212 decays to form thallium-208, the number of thallium nuclei is much less than $\frac{3}{4}N$. Suggest an explanation for these observations. [half-life of thallium = 190 s, given in the earlier part] [4]	
	Key Concepts	Radioactive half-life
	Answer	- Bismuth has a <u>half-life of one hour</u>. Therefore, <u>the number of bismuth nuclei is reduced to $\frac{1}{4}N$ after two hours</u> (two half-lives of bismuth). - The <u>number of bismuth nuclei that decayed in two hours is approximately $\frac{3}{4}N$</u>. - Since <u>one thallium nucleus is formed from the decay of one bismuth nucleus, a total of $\frac{3}{4}N$ thallium is formed in two hours</u>. - However, since thallium nuclei also decay with a <u>much shorter half-life</u> (187 s), <u>most of the thallium nuclei formed would have decayed</u>. Hence, the number of thallium nuclei is much less than $\frac{3}{4}N$.
	Remarks	Question states “suggest an explanation for these observations”. Hence you need to explain every observation mentioned. Decay constants of bismuth-212 ($1.9 \times 10^{-4} \text{ s}^{-1}$) and thallium-208 ($3.7 \times 10^{-3} \text{ s}^{-1}$) were provided in the question. Although the number of bismuth after two hours were given, it is still important to explain why this is so. Furthermore, simply stating that thallium is radioactive without any reference to the respective half-lives is insufficient to explain why the number of thallium is much less than $\frac{3}{4}N$. Note: Activity of thallium nuclei is not necessarily higher than bismuth as activity is dependent on both the number of radio nuclei and decay constant. Note: Another possible reason is that bismuth nuclei do not always undergo alpha decay to form thallium as they can also undergo beta decay to form polonium-212.

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31	2021 Paper 2 Q7(a)	Nuclear Physics
	In the original Rutherford alpha particle scattering experiment, alpha particles were fired at a thin gold foil. As a result of the experiment, the model of the atom was changed. State and explain the evidence from this experiment that changed the model of the atom in terms of its charge distribution and its mass distribution. [4]	
	Key Concepts	Rutherford experiment
	Answer	• A <u>small fraction of alpha particles undergoes large angle deflection</u>. • This implies that the <u>positive charge is concentrated in a small volume generating a strong repulsive electrostatic force</u>. • This also implies that <u>most of its mass is concentrated in a very small volume</u>. • <u>Most of the alpha particles passes through undeflected</u>. This implies that atom is <u>mostly empty space</u>.
	Remarks	At the time of Rutherford experiment in 1905, the only known elementary charged particle is electrons (protons and neutrons were discovered in 1917 and 1932, respectively) whose charge and mass were known. However, since an atom is neutral and electrons have very low mass, it is deduced that an atom must contain positive charges of large mass. First, the evidence suggested that positive charge is the cause of large deflection of alpha particles due to its large mass. Second, the low probability of large deflection suggested that this positive charge must occupy a very small volume.